Tan Wei Ru

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SUMMARY

Final-year Software Engineering student at Universiti Malaya with QA and ERP internship experience. Builds mobile and full-stack applications; explores explainable AI, geospatial data, computer vision and game development.

EDUCATION

Universiti Malaya | Bachelor of Computer Science (Software Engineering)
2023-2027 (Expected) | CGPA: 3.79/4.0
Exchange: Sejong University, South Korea | Sep-Dec 2025

EXPERIENCE

Keyway Digital Labs Sdn. Bhd. | Jul-Sep 2026
Software Engineering Intern (QA & ERP Development) | Voluntary

- Performed system-wide manual testing of KeywayEHS, covering Incidents, HIRARC, Permit-to-Work, Safety Inspection, machinery, PPE and other EHS modules.
- Authored and logged 916 test cases in a test case management system; identified approximately 130 bugs.
- Elicited requirements from HR and developed custom Odoo features using Python and XML within the Odoo framework; debugged and tested changes against requirements.

SELECTED PROJECTS

PulseCheck | Cardiac Health App | Final-Year Project (Ongoing)

Flutter, FastAPI, XGBoost, SHAP, Firebase

- Developing CVD risk prediction and PPG signal processing with a Flutter frontend; trained and compared three ML models and integrated SHAP explanations of contributing risk factors.

AegisAI | Enterprise AI Governance Gateway | Hack Attack 2026

Python, FastAPI, React, TypeScript, LiteLLM, Presidio

- Architected the backend for PII masking, multi-model routing, audit logging and reporting; contributed to the frontend in a two-person team.

FloodSense | Flood Monitoring and Emergency Response | KitaHack

Flutter, weather APIs, Google Maps

- Implemented the rainfall anomaly service, safety checklists and nearest-shelter navigation for a team flood-preparedness application.

Wick | Stress and Workload Manager

React Native (Expo), PPG, rPPG, motion detection

- Built PPG-based stress estimation, Desk Mode with rPPG and motion-based stress scoring, and Circles community features.

KeretaSewa AI | Car Rental Decision Dashboard | UMHackathon 2026

React, Node.js, Firebase, Recharts

- Built fleet KPI visualisations and an AI recommendations panel; implemented CSV ingestion with deduplication, Firestore CRUD/authentication and PDF export.

ADDITIONAL PROJECTS

MeowGotchi - Solo Godot/GDScript virtual-pet game with feeding, play and pet-state mechanics.
FitWell - Implemented trend charts for an academic health application.
PathSeeker - Implemented keyword-based job matching with Haversine distance rules.
GeoQoE Pulse (Completed Team Project) - Built a 10-stage pipeline integrating seven geospatial datasets using GIS/H3.
Rhythm Kitchen - Solo webcam cooking game using MediaPipe mesh tracking (private repository).

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, Java, JavaScript, TypeScript, Dart, GDScript, C++, SQL, HTML/CSS
Development: React, React Native (Expo), Flutter, Node.js, FastAPI, Godot; Firebase, PostgreSQL, MySQL
AI and tools: XGBoost, SHAP, scikit-learn, MediaPipe, Git, Docker, Odoo, Figma
Testing: Black-box testing, equivalence partitioning, boundary value analysis, Selenium, Appium
Languages: Mandarin (native), English, Malay; Korean (daily conversation)
Learning: Japanese (self-study and Fast Offer entry-level course), Spanish (self-study/Duolingo)

AWARDS

ASEAN GeoAI Fusion 2026 - Top 17 Finalist | Programming League National 2025 - Open Category Finalist